Abubakar Sadiq Shittu

Knoxville, Tennessee

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Research Statement

I study how users interact with security systems by measuring existing deployments, observing users, and talking to experts to figure out what works and what's broken so I can build systems that balance security with usability.

EDUCATION

University of Tennessee Knoxville, United States — 01/25–Present
Ph.D. Computer Science — GPA: 3.67/4.00

- *Passed Comprehensive Examination:* 10/26
- *Expected Graduation:* 05/28
- *Selected Coursework:* Software Engineering, Algorithms, Systems Organization, Data-Driven Security, Digital Archaeology
- *Advisor:* Dr. Scott Ruoti

Newcastle University, Newcastle upon Tyne, United Kingdom — 09/23–09/24
M.Sc. Advanced Computer Science — GPA: 4.00/4.00

- *Selected Coursework:* Machine Learning, Secure Software Development, Systems Security, Risk & Trust Management
- *Advisor:* Prof. Thomas Gross

Fountain University Osogbo, Nigeria — 03/18–06/21
B.Sc. Computer Science, First Class Honours — GPA: 4.63/5.00

- *Advisor:* Dr. Mohammed Lawal

PUBLICATIONS

Conference Proceedings

1. **Abubakar Sadiq Shittu**, Clay Shubert, John Sadik, and Scott Ruoti. "Setting Up TLS Authentication Was Hell": A Usability Study of Client Certificate Authentication. **To appear in** *Proceedings of the 2026 ACM Conference on Computer and Communications Security (CCS '26), November 15–19, 2026, The Hague, Netherlands.* [**Acceptance Rate: TBD%, TBD/TBD**]. DOI: 10.1145/3830454.3846525. Preprint: arXiv:2604.14330.

2. **Abubakar Sadiq Shittu**, John Sadik, and Scott Ruoti. "A Multi-Month Study of Git Commit Signing." **To appear in** *Proceedings of the 2026 ACM Conference on Computer and Communications Security (CCS '26), November 15–19, 2026, The Hague, Netherlands.* [**Acceptance Rate: TBD%, TBD/TBD**]. DOI: 10.1145/3830454.3846532. Preprint: arXiv:2608.29283.

Manuscripts Under Review

3. **Abubakar Sadiq Shittu**, John Sadik, Farzin Gholamrezae, and Scott Ruoti. "Analysis of Commit Signing on GitHub." Manuscript under review, 2026. Preprint: arXiv:2604.14014.

EXPERIENCE

Graduate Research Assistant — Knoxville, Tennessee, United States
University of Tennessee — 01/25–Present

- Designed and conducted mixed-methods longitudinal user studies ($n > 22$) of security systems to identify usability, functionality, and key-management barriers to adoption and correct use; analyzed qualitative self-reports, behavioral data from server logs, and standardized usability metrics (ASQ and SUS) using inferential statistics.
- Plan and conduct stakeholder interviews to examine factors influencing successful deployment of systems.
- Synthesize findings across multiple studies to identify generalizable design principles for future systems.
- Designed a stratified sampling measurement study of GitHub commit signing and built a Docker/Celery pipeline using REST and GraphQL APIs to asynchronously collect millions of user and commit records in PostgreSQL for analysis.

Information Technology Helpdesk Intern — Nigeria
Nigerian National Petroleum Corporation — 03/19–08/19

- Delivered on-demand IT support to 100+ users, diagnosing and resolving hardware and software issues daily.

AWARDS

- **Travel Grant**, *Summer School on Real-World Crypto and Privacy, Croatia* (2024).
- **Chevening Scholarship**, *UK Government-funded* (2023; approximately $70,000 in tuition, stipend, and travel support).

PEER REVIEW

Journals: *Journal of Cybersecurity* (2025, 2026).
Artifact Evaluation:

- ACM Conference on Computer and Communications Security (ACM CCS '26) (2026)
- Network and Distributed System Security Symposium (NDSS '27) (2026).

••• SKILLS

- **Programming:** Python, R, SQL, C/C++.
- **Analysis:** pandas, NumPy, scikit-learn, Matplotlib, Seaborn, ggplot2, dplyr.
- **Engineering:** PostgreSQL, Docker, Celery, Redis, ETL, API Integration.
- **Research:** Prolific, Qualtrics, NVivo, MAXQDA, LaTeX.
- **Tools:** Git, SonarQube, Wireshark, Linux/Unix, Windows.

AFFILIATIONS & COMMUNITY

- **IEEE Computer Society:** Graduate Student Member
- **Tennessee Extension:** Volunteer Participant